# PrismDM: An Accurate and Efficient Framework for Privacy-Preserving Diffusion Model Sampling

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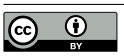
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## Abstract

Diffusion models (DMs) have gained significant attention for their powerful image synthesis capabilities. However, the applications of DMs may expose sensitive user inputs, thereby raising substantial privacy concerns. Existing privacy-preserving methods offer rigorous privacy guarantees but incur excessive computational and communication overhead. While several acceleration techniques have been developed, directly applying them to DMs often results in severe image quality degradation. To the best of our knowledge, we are the first to identify this degradation and trace its root cause to the coupling between the acceleration-induced bias and the iterative nature of DM sampling. Based on this insight, we propose PrismDM, the first secure DM sampling framework that jointly considers efficiency and accuracy. We propose an accuracy-aware acceleration method to mitigate the bias via multi-objective optimization, along with a feature alignment algorithm that stabilizes the sampling trajectory, thereby improving both sample efficiency and quality. Compared to state-of-the-art methods, PrismDM improves sample quality by 1.56× ~ 14.66×, reduces inference time by 1.07× ~ 3.15×, and minimizes communication overhead by 1.18× ~ 4.20×.

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## CCS Concepts

• **Security and privacy → Privacy-preserving protocols**; • **Computing methodologies → Computer vision**; **Machine learning**.

## Keywords

Privacy-Preserving Machine Learning, Diffusion Model, Multi-Party Computation

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## 1 Introduction

Diffusion Models (DMs) [12, 34, 36] have been widely adopted in websites and online applications due to their powerful generation capabilities [1, 9, 13]. However, these services inherently involve processing user data in untrusted or user-unverifiable execution environments, thereby posing substantial privacy risks [5, 10, 43]. For instance, DM-powered online image editing typically requires users to upload their private images and prompts, which may be exposed to unauthorized access or misuse by adversarial entities [19].

To address those concerns, secure Multi-Party Computation (MPC) has been extensively studied [16, 25, 30, 42]. Despite offering rigorous privacy guarantees, MPC suffers from significant computational overhead [2, 31]. Prior work has explored various ways to

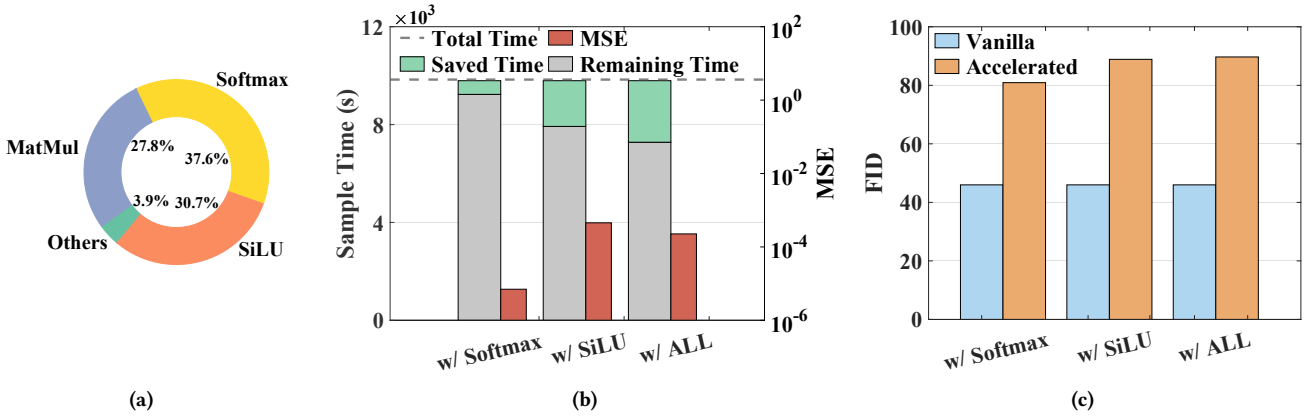



**Figure 1: (a) DM sampling time breakdown by operator under MPC. (b) Sampling time reduction and induced bias magnitude for different accelerated operators ("w/" denotes with and "ALL" means accelerating both SiLU & Softmax, "MSE" denotes mean square error and measures magnitude of bias). (c) Image quality degradation of accelerated models (Vanilla denotes models without acceleration). Higher Fréchet Inception Distance (FID) [8] indicates lower quality and accuracy.**

mitigate this overhead, such as improving low-level computation primitives [22, 24] and simplifying inference pipelines [20]. Among these, approximating nonlinear operators with computationally efficient polynomials has emerged as a mainstream strategy. This approach sacrifices minor accuracy for faster inference and has proven effective in discriminative and language models [6, 26, 32].

Such strategy has been further applied to DMs. A representative example is CipherDM [44], the state-of-the-art (SOTA) method in secure DM sampling. It proposes the first secure DM sampling protocol and employs piecewise polynomial approximation to reduce inference cost and achieve high efficiency. However, its strategy primarily targets sampling efficiency, while neglecting generation quality. As shown in Fig. 1, even slight acceleration-induced bias can be greatly amplified, leading to substantial quality degradation and rendering secure DM sampling infeasible.

In this work, we take the first step towards achieving secure DM sampling under efficiency and quality constraints. We begin by systematically analyzing the bias amplification and attribute the compromised quality to two key challenges. First, existing acceleration schemes rely on manually designed approximation methods, which lack joint considerations of efficiency and accuracy. Second, the iterative nature of DM sampling amplifies small biases in intermediate feature maps, which accumulate over iterations and lead to structural collapse, ultimately degrading sample quality.

To address these challenges, we propose PrismDM, the first accurate and efficient privacy-preserving DM sampling framework based on MPC. It integrates multi-objective acceleration to balance efficiency and accuracy, and trajectory regularization to preserve structural consistency, thereby enabling secure, efficient, and high-quality DM sampling. Our contributions are summarized as follows:

**Systematic Quality Degradation Analysis.** To the best of our knowledge, we are the first to identify severe quality degradation in accelerated secure DM sampling. This degradation critically hinders practical secure DM sampling, which requires both high quality and efficiency. Furthermore, we systematically analyze the degradation and trace its root causes to the coupling between acceleration-induced bias and iterative sampling process, offering core insights that guide the design of secure and practical DM sampling protocol.

**Secure and Accurate Sampling Framework.** To address the limitations of existing acceleration methods that lead to quality degradation, we introduce an accuracy-aware acceleration scheme and a feature alignment training algorithm, which together form PrismDM, a secure DM sampling framework that achieves high sample quality with low inference cost.

**Generalizable and Versatile Secure DM.** We evaluate PrismDM's effectiveness across diverse and representative DMs, including DDPM, DDIM, and StableDiffusion. Compared to the SOTA method CipherDM, our method achieves an accuracy improvement of 1.56× to 14.66×, minimizes the inference time by 1.07× to 3.15×, and reduces the communication overhead by 1.18× to 4.20×.

## 2 Problem Formulation and Preliminary

### 2.1 Problem Formulation

We formulate the secure DM sampling problem with three types of participants: a client holding a private input $x$, a model developer owning a vanilla (i.e., unaccelerated) DM $M$, and a set of computation parties $P_i$ that jointly execute the secure sampling protocol. The primary goal is to allow the client to obtain the output $y = M(x)$, while ensuring that $x$ and $y$ remain private to the client, the model $M$ remains private to the model developer, and the computation parties $P_i$, as well as any other entities, learn nothing about $x, y, M$ during sampling process. In addition, practical secure DM sampling requires generating high-quality results while achieving high efficiency, with efficiency measured by inference cost. Inference cost refers to total inference time and amount of data transmitted among MPC parties during the secure sampling process.

### 2.2 Cryptographic Scheme

**MPC Framework.** We adopt secure Multi-Party Computation (MPC) as the core cryptographic framework in this work, due

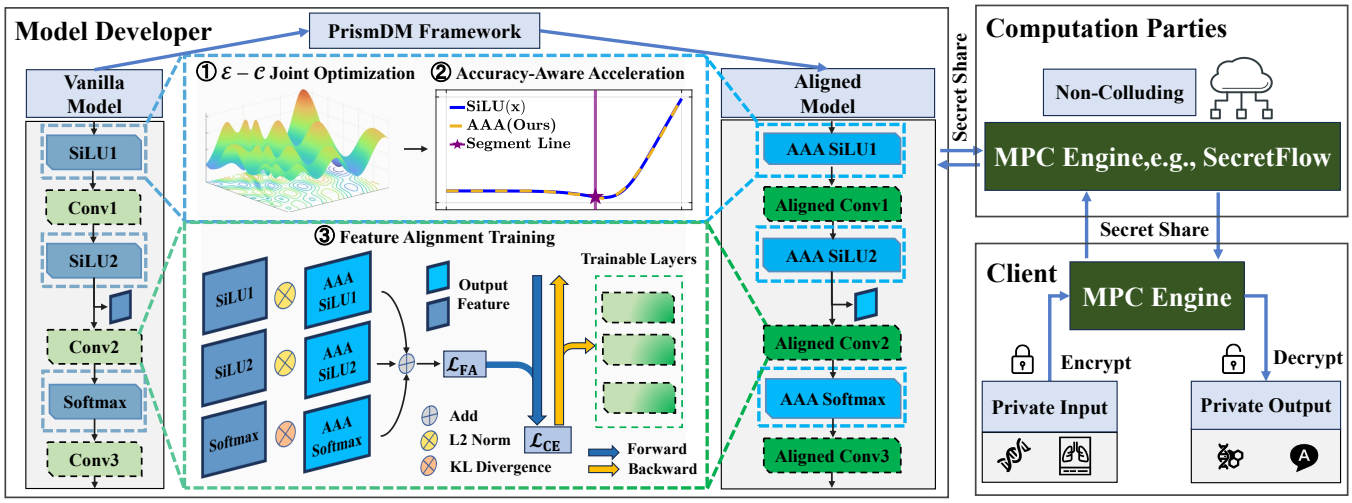



**Figure 2: Overview of PrismDM framework for privacy-preserving DM sampling; $\mathcal{E}, C$ denote accuracy and efficiency terms in multi-objective optimizations respectively, "AAA" denotes Accuracy-Aware Acceleration.**

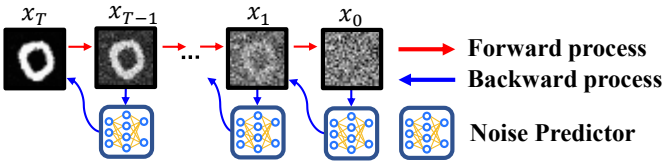



**Figure 3: Schematic of DM sampling process.**

to its favorable trade-offs between privacy, efficiency, and accuracy [3, 41]. MPC operates by secret-sharing inputs among multiple non-colluding computation parties, who then jointly perform computation without revealing the original inputs. Specifically, we use the widely adopted three-party framework ABY3 [27], which employs a 2-out-of-3 secret sharing scheme and offers superior efficiency compared to two-party protocols [28].

**2-out-of-3 Secret Sharing.** In 2-out-of-3 secret sharing, a secret $x$ in ring $\mathbb{Z}_{2^l}$ is divided into shares $[\![x]\!] = x_0 + x_1 + x_2$, where $l$ is the bit length and is set to 18 in this work. These shares are distributed among three computation parties $P_i$, where each $P_i$ holds $(x_i, x_{(i+1) \bmod 3})$. We list basic cryptographic notations of MPC protocols in Tab. 1 and summarize computation primitives used to form composite protocols (e.g., full inference protocol) below.

**Addition.** For secrets $[\![x]\!]$, $[\![y]\!]$ and public constants $c_1, c_2, c_3$, linear additions are computed as:

$$[\![c_1 x + c_2 y + c_3]\!] = c_1[\![x]\!] + c_2[\![y]\!] + c_3. \tag{1}$$

Each party $P_i$ locally computes $c_1 x_i + c_2 y_i + \delta_{i0} c_3$, preserving additive homomorphism, where $\delta_{i0}$ is 1 for $i = 0$ and 0 otherwise.

**Multiplication.** Given secret-shared inputs $[\![x]\!]$ and $[\![y]\!]$, the multiplication protocol $\Pi_{\text{Mul}}$ computes a secret sharing of $x \cdot y$. Concretely, the plaintext product can be decomposed as

$$x \cdot y = \sum_{i=0}^{2} x_i \cdot \sum_{j=0}^{2} y_j = \sum_{0 \le i,j \le 2} x_i \cdot y_j = w_0 + w_1 + w_2, \tag{2}$$

**Table 1: Basic notations for MPC protocols.**

| Notation | Meaning |
|---|---|
| $\mathbb{Z}_{2^\ell}$ | ring modulo $2^\ell$ |
| $P_i$ | computation party $i$ in 3PC |
| $[\![\cdot]\!]$ | arithmetic sharing over $\mathbb{Z}_{2^\ell}$ |
| $[\![\cdot]\!]^B$ | Boolean sharing over $\mathbb{Z}_2$ |
| $\mathbf{x}$ | vector |
| $x$ | scalar |
| $x^i$ | the $i$-th power of scalar $x$ |
| $\alpha_i$ | random mask precomputed by $P_i$ |
| $\oplus$ | XOR operation |

**Table 2: Underlying black-box MPC protocols.**

| Protocol | Functionality | Description |
|---|---|---|
| $\Pi_{\text{LT}}$ | $z = 1\{x < y\}$ | less-than comparison |
| $\Pi_{\text{Max}}$ | $z = \max(\mathbf{x})$ | maximum |
| $\Pi_{\text{Recip}}$ | $z = 1/x$ | reciprocal |
| $\Pi_{\text{Square}}$ | $z = x^2$ | square |
| $\Pi_{\text{Mul}_{\text{BA}}}$ | $z = b \cdot x$ | Boolean-arithmetic multiplication |

where each party $P_i$ locally computes $w_i = x_i \cdot y_i + x_{i+1} \cdot y_i + x_i \cdot y_{i+1}$. Each party $P_i$ then sends the masked value $w_i' = w_i - \alpha_i$ to $P_{i+1}$, where the masks $\alpha_i$ satisfy $\sum_i \alpha_i = 0$ and are generated in the setup phase. The protocol finally forms $\Pi_{\text{Mul}}([\![x]\!], [\![y]\!])$ with the secret sharing $\{(w_0', w_1'), (w_1', w_2'), (w_2', w_0')\}$.

**Underlying Protocols.** PrismDM leverages SOTA computation primitives (e.g., comparison, truncation) implemented in SecretFlow [25], treating them as black-box components. Input/output specifications for these protocols are listed in Tab. 2.

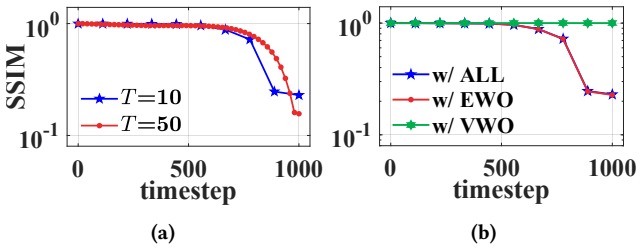



**Figure 4: Different factors' impact on sample quality (SSIM). (a) Sampling steps ($T$). (b) Different accelerated nonlinear operators ("w/" denotes acceleration with and "ALL" means accelerating both EWO & VWO).**

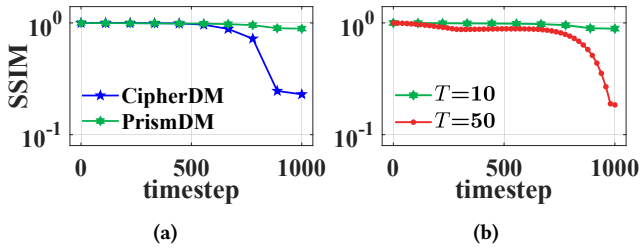



**Figure 5: (a) Quality trend under different acceleration methods with $T$=10, "PrismDM" refers to using accuracy-aware acceleration. (b) Quality trend with accuracy-aware acceleration scheme under different $T$.**

## 2.3 Threat Model and Security Proof

**Threat Model.** We consider an honest-but-curious (HBC) adversary that corrupts at most one out of three computation parties. The adversary strictly follows MPC protocols but attempts to infer private data from observed messages. The computation parties are assumed to be *non-colluding*, following standard assumptions in prior MPC works [27, 29]. This setting is practically feasible, as there are multiple independent cloud service providers, each capable of offering computational resources (e.g., AWS, GCP, Azure). In such deployments, collusion among the computation parties is generally unlikely due to their separate administrative and compliance boundaries. Implementation-level side-channel threats, such as memory access pattern leakage, are orthogonal to the threat model considered here. These risks can be mitigated through standard system-level protections and careful deployment.

**Security Proof.** All primitives (addition, multiplication, underlying protocols) in our secure sampling protocol satisfy HBC security under the ring, as proven in ABY3 [27]. For composite protocols, security follows the *modular composition theorem* [4]. Since each sub-protocol (e.g., add, truncation) is HBC secure and their sequential composition does not expose intermediate shares, the overall protocol preserves privacy.

## 2.4 DM Sampling and Image Quality Evaluation Metrics

Diffusion models generate data by iteratively transforming random noise $x_0$ into structured outputs $x_T$ through a forward and reverse process, where $T$ denotes total number of sampling steps [37]. In the forward process, noise is gradually added to the data, while the reverse process denoises it using the inference results of a learned noise predictor, as illustrated in Fig. 3. Here, $x_t$ denotes the intermediate result at timestep $t$, and $x_T$ is the final output. To evaluate sample quality, we adopt **Fréchet Inception Distance (FID)** [8], a standard metric assessing fidelity and diversity of generated images. FID is computed as:

$$\mathrm{FID} = \|\mu_{\mathrm{r}} - \mu_{\mathrm{g}}\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{1/2}\right), \tag{3}$$

where $\|\cdot\|_2$ denotes the L2 norm and Tr is the trace operator. $\mu_r$, $\Sigma_r$ and $\mu_g$, $\Sigma_g$ are the mean and covariance of features extracted from real and generated images via a pre-trained model. Lower FID indicates better sample quality.

# 3 Method

## 3.1 Overview

PrismDM framework consists of two stages to enable practical secure DM sampling: an offline training phase conducted by model developer, and an online secure inference phase executed by MPC computation parties, as shown in Fig. 2.

**Training Phase.** To address accuracy bottleneck, we introduce an accuracy-aware acceleration scheme that replaces nonlinear operators in the vanilla model $M$ with polynomial surrogates, balancing acceleration accuracy and computational cost. The surrogate model $\widehat{M}$ is then further fine-tuned to the aligned model $\widetilde{M}$ using the feature alignment algorithm, which regularizes intermediate features during sampling to mitigate accumulated feature bias.

**Inference Phase.** The aligned model and client inputs are secret-shared among the MPC parties, who jointly execute the secure sampling protocol over encrypted inputs. After secure inference, the client reconstructs the output from secret shares, ensuring end-to-end privacy for both data and model.

## 3.2 Bias Amplification Factors

We start with identifying two key factors related to bias amplification: the number of sampling steps $T$ and the magnitude of acceleration-induced bias. To assess sample quality of the surrogate model, we measure its output similarity to vanilla model across timesteps. We adopt Structural Similarity Index Measure (SSIM) [39] as similarity metric due to its alignment with human perception of image quality. A higher SSIM score indicates greater similarity to vanilla output, and thus serves as a proxy for sample quality.

Furthermore, since various nonlinear operators' approximations compound the magnitude of bias, we categorize them into two groups: Element-wise Operators (EWO), including ReLU, SiLU, and Mish, and Vector-wise Operators (VWO) , such as Softmax. Fig. 4a shows the similarity trends under different $T$. When $t < 600$, the similarity remains high, primarily because both models' outputs $x_t$ at this stage are heavily dominated by noise. However, a significant similarity decline is observed when $t \geq 600$, with higher $T$ aggravating the degradation. This phenomenon aligns with the intuition that more inference cycles tend to amplify bias more severely.

Meanwhile, we assess impact of bias magnitude on quality degradation by only accelerating specific type of nonlinear operators. As shown in Fig. 4b, when $t \geq 600$, accelerating EWO leads to the

most severe SSIM degradation. Given that EWOs greatly outnumber VWOs, and the acceleration bias of EWO is significantly higher, it is evident that lower approximation precision and more approximated operators compound the magnitude of bias amplification.

### 3.3 Accuracy-Aware Acceleration

Prior analysis reveals that acceleration-induced bias plays a critical role in quality degradation. Traditional acceleration methods, such as BOLT [32], Bumblebee [24], CipherDM [44] and MPCFormer [18], rely on either piecewise approximations with heuristic tuning or global approximations with retraining [7, 23]. These designs lack unified modeling of accuracy and cost, and are inadequate for DMs where iterative sampling amplifies the bias. In contrast, we cast acceleration as a multi-objective optimization over interval boundaries, polynomial degrees, and candidate functions to create accurate and MPC-efficient surrogate operators.

Let $f(x)$ be the operator to accelerate, and let $\mathcal{N}$ denote the distribution of $x$. We divide input domain into $K$ intervals with boundaries $\{z_i\}_{i=1}^{K-1}$. For each interval $[z_{i-1}, z_i]$, $f(x)$ is approximated using polynomials, as non-polynomial operators (e.g., wavelet operators) incur higher MPC costs. The approximation is parameterized by a polynomial degree $L_i \in \mathbb{Z}^+$ and a sparsity pattern $j_i$ indicating included monomial terms, yielding a polynomial $p_{L_i,j_i}(x)$.

The accuracy loss $\mathcal{E}$ introduced by acceleration consists of approximation error and continuity penalty. The approximation error $\mathcal{E}_{\text{app}}$ measures the functional deviation between the original operator $f(x)$ and its polynomial surrogate $p(x)$:

$$\mathcal{E}_{\text{app}} = \mathbb{E}_{x\sim\mathcal{N}}\left[(f(x) - p(x))^2\right], \tag{4}$$

with the continuity penalty $\mathcal{E}_{\text{cont}}$ defined as:

$$\mathcal{E}_{\text{cont}} = \sum_{i=1}^{K-1} \left(p_{L_i,j_i}(z_i) - p_{L_{i+1},j_{i+1}}(z_i)\right)^2. \tag{5}$$

For inference cost under secure computation, since all branches must be evaluated to avoid leakage of control flow, we model the inference cost as:

$$C = \sum_{i=1}^{K} \tau(p_{L_i,j_i}), \tag{6}$$

where $\tau(\cdot)$ denotes the runtime cost of securely evaluating the polynomial in the MPC environment. In practice, we instantiate $\tau(\cdot)$ using the average inference time measured under selected MPC protocol and settings.

Putting accuracy and efficiency together, the final optimization objective is formulated as:

$$\min_{\{z_i\},\{L_i\},\{j_i\}} \quad \alpha \cdot (\mathcal{E}_{\text{app}} + \Lambda \cdot \mathcal{E}_{\text{cont}}) + \beta \cdot C, \tag{7}$$

where $\Lambda \in \mathbb{R}^+$ controls smoothness at interval boundaries. $\alpha, \beta \in \mathbb{R}^+$ balance approximation accuracy and inference efficiency by compensating for scale differences.

**Acceleration Scheme Application.** To validate our acceleration framework, we apply it to both EWOs and VWOs. For each operator, we construct a candidate space of $z_i, L_i, j_i$ on input distributions and approximation considerations, and select the optimal candidate by optimizing the objective. For hyperparameter selection, we set $\Lambda$=100 to avoid discontinuous polynomial surrogates and distortion in generated images and select $\alpha, \beta$ via a grid search, choosing the configuration that achieves low inference cost and acceptable quality.

**Comprehensive Analysis on Accuracy.** To verify the effectiveness of our scheme, we measure similarity trends between vanilla and surrogate models' outputs under different acceleration methods. As shown in Fig. 5a, our scheme maintains high sample quality under low $T$, while CipherDM's scheme exhibits a severe accuracy drop. Moreover, Fig. 5b shows that the acceleration scheme still faces a severe quality drop under high $T$, indicating that simply increasing fitting precision cannot mitigate the influence of $T$ on bias accumulation.

### 3.4 Feature Alignment Algorithm

Given the necessity of acceleration methods in MPC, further suppressing amplification of acceleration bias is critical. To achieve this, we propose a feature alignment algorithm that trains the vanilla model with selected acceleration method. The algorithm aims to regularize the intermediate features of nonlinear operator layers in the surrogate model with those in the vanilla model throughout the sampling process.

We consider the vanilla model as $M = M_1, M_2, \ldots, M_n$, where $M_i$ represents the $i$-th layer inside the model. Let $\widehat{M}$ denote the surrogate model for training, the feature alignment loss at timestep $t$ is defined as:

$$\mathcal{L}_{\text{FA}}(t) = \lambda_{\text{EWO}}\mathcal{L}_{\text{EWO}}(t) + \lambda_{\text{VWO}}\mathcal{L}_{\text{VWO}}(t), \tag{8}$$

$$\begin{cases} \mathcal{L}_{\text{EWO}}(t) = \sum_i \|M_{\text{EWO}_i}(t) - \widehat{M}_{\text{EWO}_i}(t)\|_2^2, \\ \mathcal{L}_{\text{VWO}}(t) = \sum_i D_{\text{KL}}(M_{\text{VWO}_i}(t) | \widehat{M}_{\text{VWO}_i}(t)), \end{cases} \tag{9}$$

where $M_{\text{EWO}_i}(t)$,$M_{\text{VWO}_i}(t)$ denote the outputs of layers with EWO and VWO, respectively.

Based on unique features of operators, we apply the L2 norm to the EWO layers to measure element-wise distance and the KL divergence $D_{\text{KL}}$ to the VWO layers to quantify distribution distance [15]. The pseudocode for the feature alignment algorithm is provided in Algorithm 1. $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss, and $\lambda_{\text{FA}}$ is the hyperparameter controlling the feature alignment loss. The training hyperparameters $\lambda_{\text{EWO}}, \lambda_{\text{VWO}}, \lambda_{\text{FA}}$ are tuned via a hyperparameter sweep based on the quality of generated images.

## 4 Experiment

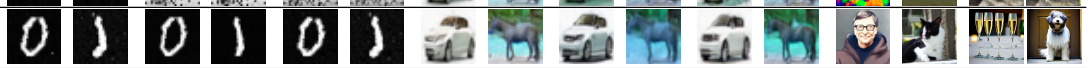
### 4.1 Experimental Setup

**Training Configuration.** We evaluate PrismDM on three representative DMs: DDPM [12], DDIM [35], and StableDiffusion V1-4. For DDPM and DDIM, both vanilla and aligned models are trained with three EWOs (ReLU, SiLU, and Mish) on MNIST-01 [17] and CIFAR-10 [14], where models for more complex datasets use more intricate architectures with additional nonlinear operators, increasing inference cost. For StableDiffusion [34], fine-tuning is conducted on lambdalabs/naruto-blip-caption (abbreviated as Naruto), as recommended by Diffusers [38], and COCO-2017 [21]. For feature alignment training consumption, DDPM/DDIM takes about 1 hour on MNIST and 3 hours on CIFAR, while StableDiffusion costs about 6 hours. Experiments are conducted on a server with an A800-80GB GPU, Intel(R) Xeon(R) Gold 6348 CPU, and 100GB RAM.

**Table 3: FIDs of different DMs. FID scores are computed using 12,000 generated images per DM. "w/o FA" denotes models without feature alignment training. "Accuracy ↑ w/ PDM" indicates $\text{FID}_{\text{CipherDM}}/\text{FID}_{\text{PrismDM}}$. "__" indicates FIDs of vanilla (i.e. unaccelerated) DMs.**

| Models | DDIM DM w/ $T$=50 | | | | | | StableDiffusion w/ $T$=50 | |
|---|---|---|---|---|---|---|---|---|
| Datasets | MNIST-01 | | | CIFAR-10 | | | Naruto | COCO |
| | ReLU | SiLU | Mish | ReLU | SiLU | Mish | | |
| SecretFlow | 123.49 | 45.98 | 68.56 | 61.22 | 48.99 | 56.86 | 130.63 | 25.27 |
| CipherDM | 114.72 | 430.56 | 395.95 | 342.09 | 341.44 | 319.33 | 363.78 | 419.06 |
| MPCFormer | 81.77 | 394.58 | 327.44 | 54.48 | 100.5 | 108.61 | 308.14 | 214.62 |
| PrismDM w/o FA | 107.99 | 360.22 | 403.94 | 61.22 | 252.93 | 317.41 | 409.98 | 278.47 |
| PrismDM | **73.52** | **48.46** | **73.81** | **46.23** | **67.31** | **57.52** | **124.63** | **28.59** |
| Accuracy ↑ w/ PDM | 1.56× | 8.88× | 5.36× | 7.39× | 5.07× | 5.55× | 2.92× | 14.66× |

**Table 4: Visual examples of images generated by different DMs. SecretFlow denotes vanilla (i.e. unaccelerated) DMs.**

| Models | DDIM DM w/ $T$=50 | | | | | | StableDiffusion w/ $T$=50 | |
|---|---|---|---|---|---|---|---|---|
| Datasets | MNIST-01 | | | CIFAR-10 | | | Naruto | COCO |
| | ReLU | SiLU | Mish | ReLU | SiLU | Mish | | |
| SecretFlow |  | | | | | | | |
| CipherDM |  | | | | | | | |
| MPCFormer |  | | | | | | | |
| PrismDM |  | | | | | | | |

**Algorithm 1** Feature Alignment Algorithm

**Input:** Vanilla model $M$, surrogate model $\widehat{M}$, training dataset $\mathcal{D}_T$
**Output:** Aligned model $\widetilde{M}$

Initialize $\widetilde{M}$ with surrogate model $\widehat{M}$
**while** not converged **do**
  Sample training data batch $(X_T, Y_T)$ from $\mathcal{D}_T$
  **for** each sample $i$ in the batch **do**
    Randomly sample timestep $t_i \in \{1, 2, \ldots, T\}$
    Generate noise $\epsilon_{t_i}$ from the forward diffusion process
    Model predict noise $\hat{\epsilon}_{t_i} = \widetilde{M}(X_{T_i}, Y_{T_i}, t_i)$
    Compute losses: $\mathcal{L}_{\text{CE}} = \text{CE}(\epsilon_{t_i}, \hat{\epsilon}_{t_i})$
    $\mathcal{L}_{\text{EWO}} = \sum_i \|M_{\text{EWO}_i}(t_i) - \widetilde{M}_{\text{EWO}_i}(t_i)\|_2^2$
    $\mathcal{L}_{\text{VWO}} = \sum_i D_{\text{KL}}(M_{\text{VWO}_i}(t_i) | \widetilde{M}_{\text{VWO}_i}(t_i))$
    $\mathcal{L}_{\text{FA}} = \lambda_{\text{EWO}} \cdot \mathcal{L}_{\text{EWO}} + \lambda_{\text{VWO}} \cdot \mathcal{L}_{\text{VWO}}$
    Update $(\widetilde{M}, \mathcal{L}_{\text{CE}} + \lambda_{\text{FA}} \cdot \mathcal{L}_{\text{FA}})$
  **end for**
**end while**

**MPC System Setup.** Following CipherDM, we evaluate PrismDM's efficiency built upon the widely used MPC framework SecretFlow [25] and use it as baseline to denote vanilla (i.e. unaccelerated) model. We use the Linux `tc` tool to simulate LAN with a bandwidth of 1Gbps. Experiments are carried out on a server equipped with an Intel(R) Xeon(R) Gold 5318S CPU and 1TB RAM.

**Evaluation Baselines.** We consider three baselines in our evaluation: the vanilla DM without acceleration (denoted as SecretFlow), and two typical acceleration strategies for MPC inference, CipherDM, and MPCFormer. CipherDM and MPCFormer are algorithm-level acceleration methods rather than tied to particular models and can be reproduced from their papers and open-source code. We faithfully reproduce these baselines following their original papers and official code repositories, apply all methods to the same pretrained vanilla DMs, and evaluate them under same settings for fair comparison.

## 4.2 Accuracy & Image Quality

In this section, we evaluate the effectiveness of PrismDM in improving sample quality. As shown in Tab. 3, our approach significantly enhances the quality of generated images, with improvements ranging from 1.56× to 14.66×. This substantial boost demonstrates that the proposed acceleration strategy, together with the training algorithm, effectively mitigates the bias amplified by iterative sampling.

Furthermore, although the vanilla models (SecretFlow) rely on the original nonlinear operators and are thus intuitively expected to achieve higher sample quality, the aligned models (PrismDM) outperform them in several settings, including DDIM models with

**Table 5: Inference cost on DDIM & DDPM. "Time" is measured in seconds and "Comm." denotes communication measured in gigabytes. "†" denotes results that are fast but lack quality. "Local" denotes running all MPC parties on a single server to measure pure computation cost, while "LAN" denotes running MPC parties on separate servers in a local area network to evaluate network impact.**

| **DDIM** | MNIST-01 | | | | | | CIFAR-10 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Time** | ReLU | | SiLU | | Mish | | ReLU | | SiLU | | Mish | |
| | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN |
| SecretFlow | 251 | 472 | 343 | 621 | 394 | 748 | 935 | 1432 | 2066 | 4243 | 2844 | 6655 |
| CipherDM | 240 | 451 | 315 | 552 | 320 | 551 | 909 | 1387 | 1483 | 2380 | 1488 | 2390 |
| MPCFormer | 151† | 183† | 175† | 224† | 177† | 211† | 743† | 1043† | 961† | 1472† | 968† | 1465† |
| PrismDM | **215** | **395** | **261** | **464** | **260** | **463** | **884** | **1341** | **1314** | **2101** | **1329** | **2115** |
| **Improve ↑** | 1.17× | 1.20× | 1.31× | 1.34× | 1.52× | 1.62× | 1.06× | 1.07× | 1.57× | 2.02× | 2.14× | 3.15× |
| **Comm.** | ReLU | | SiLU | | Mish | | ReLU | | SiLU | | Mish | |
| | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN |
| SecretFlow | 9.13 | 9.10 | 12.95 | 12.91 | 15.84 | 15.85 | 27.28 | 27.38 | 115.60 | 115.63 | 183.23 | 183.50 |
| CipherDM | 8.43 | 9.11 | 10.23 | 10.25 | 10.25 | 10.27 | 25.42 | 25.33 | 53.57 | 53.61 | 53.62 | 53.67 |
| MPCFormer | 1.82† | 1.79† | 2.63† | 2.65† | 2.62† | 2.66† | 16.22† | 16.21† | 26.57† | 26.59† | 26.58† | 26.59† |
| PrismDM | **6.64** | **6.51** | **7.43** | **7.43** | **7.44** | **7.43** | **23.09** | **23.12** | **43.59** | **43.79** | **43.74** | **43.71** |
| **Improve ↑** | 1.38× | 1.40× | 1.74× | 1.74× | 2.13× | 2.13× | 1.18× | 1.18× | 2.65× | 2.64× | 4.19× | 4.20× |
| **DDPM** | MNIST-01 | | | | | | CIFAR-10 | | | | | |
| **Time** | ReLU | | SiLU | | Mish | | ReLU | | SiLU | | Mish | |
| | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN |
| SecretFlow | 9373 | 12160 | 11310 | 15260 | 12680 | 17614 | 31800 | 40487 | 49655 | 94399 | 61589 | 141657 |
| CipherDM | 9022 | 11766 | 11014 | 14691 | 11921 | 14894 | 30589 | 38768 | 42954 | 58594 | 43258 | 76592 |
| MPCFormer | 5740† | 5931† | 5159† | 5670† | 5149† | 5670† | 27840† | 31642† | 30610† | 43558† | 30981† | 44498† |
| PrismDM | **8145** | **9913** | **8587** | **11082** | **9948** | **12133** | **29630** | **37674** | **39648** | **54978** | **39365** | **57559** |
| **Improve ↑** | 1.15× | 1.23× | 1.32× | 1.38× | 1.27× | 1.45× | 1.07× | 1.07× | 1.25× | 1.72× | 1.56× | 2.46× |
| **Comm.** | ReLU | | SiLU | | Mish | | ReLU | | SiLU | | Mish | |
| | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN | Local | LAN |
| SecretFlow | 182.20 | 182.36 | 258.52 | 258.66 | 317.05 | 317.03 | 547.18 | 547.27 | 2312.73 | 2312.93 | 3670.35 | 3670.81 |
| CipherDM | 161.53 | 160.29 | 202.91 | 204.57 | 204.97 | 204.86 | 498.67 | 498.17 | 1069.80 | 1069.93 | 1087.83 | 1087.91 |
| MPCFormer | 36.16† | 36.03† | 44.99† | 44.98† | 45.02† | 44.92† | 323.73† | 323.49† | 531.18† | 530.91† | 530.88† | 530.81† |
| PrismDM | **131.47** | **131.23** | **144.42** | **144.47** | **144.50** | **144.51** | **462.56** | **462.41** | **784.02** | **785.82** | **873.05** | **873.58** |
| **Improve ↑** | 1.38× | 1.39× | 1.79× | 1.79× | 2.19× | 2.19× | 1.18× | 1.18× | 2.95× | 2.94× | 4.20× | 4.20× |

ReLU activation. This can be primarily attributed to the fact that PrismDM's polynomial surrogates preserve useful negative activation values that are clipped by ReLU, thereby leading to richer intermediate representations. In addition, features from the vanilla models provide smoothed representations of outlier samples during feature alignment training [33], which helps prevent the aligned models from overfitting. Meanwhile, even though MPCFormer achieves moderate accuracy on simple models (ReLU-based DDIMs), its inaccurate acceleration scheme blocks it from being practical in more advanced and powerful DMs such as DDIMs on SiLU and StableDiffusion, as indicated by its high FID scores.

To further illustrate the impact of different acceleration methods on image quality, we present visual examples of the generated images. As shown in Tab. 4, acceleration-induced bias results in more severe distortions in DMs with complex nonlinear operators (e.g., SiLU, Mish) and advanced model architectures (e.g., Stable Diffusion). In simpler datasets with limited color and structural diversity, such as MNIST-01, this bias often results in entirely noisy and visually meaningless samples. In contrast, for more complex and color-diverse datasets like CIFAR-10 and COCO, the bias primarily degrades the model's ability to generate fine-grained details, thereby compromising structural coherence and producing outputs with irregular textures and indistinct features.

We further evaluate different methods on two key capabilities of text-to-image generation: prompt-image alignment and alignment with human preferences, using the widely adopted CLIPScore [11] and ImageReward [40] as evaluation metrics, respectively. As shown in Tab. 6, PrismDM consistently outperforms the other acceleration

**Table 6: Text-to-Image Ability Evaluation on COCO. "__" indicates scores of vanilla (i.e. unaccelerated) DMs.**

| Metrics (↑) | Methods | | | |
|---|---|---|---|---|
| | SecretFlow | CipherDM | MPCFormer | PrismDM |
| CLIPScore | 31.36 | 19.75 | 23.72 | **31.07** |
| ImageReward | 0.18 | -2.27 | -2.04 | **0.08** |

methods on both metrics, demonstrating its effectiveness in preserving semantic consistency and alignment with human preferences during accelerated secure sampling.

## 4.3 Inference Overhead

In this section, we evaluate the effectiveness of PrismDM in improving sampling efficiency. The metric **Improve** ↑ denotes the time and communication reduction ratios achieved by PrismDM compared to model without acceleration (SecretFlow). For DDIM and DDPM, the sampling steps are set to 50 and 1000 to ensure fair and meaningful sampling, respectively. For StableDiffusion, we focus on measuring the computational cost of its core component, the U-Net model.

As shown in Tab. 5 and Tab. 7, our method significantly reduces inference time and communication cost. It outperforms the previous SOTA method CipherDM, achieving 1.06× to 3.15× speedup and 1.18× to 4.20× reduction in communication overhead on DDIM and StableDiffusion, demonstrating broad applicability. In the case of DDPM, the proposed method reduces communication costs across different models, although its improvements in inference time are limited. This limitation is less critical, since DDIM is capable of producing comparable-quality samples with fewer steps. On the other hand, while MPCFormer achieves the lowest computational cost, its sample quality is severely compromised and becomes meaningless, as most of its FID scores exceed thresholds reflecting practical generation quality.

In summary, across diverse and representative DMs, PrismDM achieves minimal inference time and communication overhead without sacrificing generation quality, enabling efficient and accurate privacy-preserving DM sampling.

## 4.4 Ablation Study

**Generation Ability Sensitivity.** We investigate the effect of different training losses on sample quality using four DDIM models fine-tuned on CIFAR-10 with controlled hyperparameters. All models employ the proposed accuracy-aware acceleration method. As shown in Tab. 8, all fine-tuned models outperform the untrained surrogate model. Among them, SettingA (with both $\mathcal{L}_{\text{VWO}}$ and $\mathcal{L}_{\text{EWO}}$) achieves the best FID, while SettingB and SettingC, each lacking one loss, perform less effective, indicating that both terms are essential for enhancing image quality under acceleration.

To further understand this interaction, we present the training loss trajectories in Fig. 6. We observe that $\mathcal{L}_{\text{EWO}}$ and $\mathcal{L}_{\text{VWO}}$ losses decrease when explicitly applied, highlighting their roles in minimizing distortion. Interestingly, both also decline in SettingD where neither $\mathcal{L}_{\text{EWO}}$ nor $\mathcal{L}_{\text{VWO}}$ is included, implying implicit alignment via optimization solely on $\mathcal{L}_{\text{CE}}$, though with inferior FID.

**Table 7: Inference cost on StableDiffusion (1 Step). "Comm." and "†" follow the same definitions as in Tab. 5.**

| Methods | Time | | Comm. | |
|---|---|---|---|---|
| | Local | LAN | Local | LAN |
| SecretFlow | 12377 | 18318 | 1241.45 | 1241.74 |
| CipherDM | 10113 | 15574 | 1188.12 | 1187.92 |
| MPCFormer | 5204† | 7441† | 235.72† | 235.87† |
| PrismDM | **6825** | **10169** | **762.25** | **762.37** |
| **Improve** ↑ | 1.81× | 1.80× | 1.63× | 1.63× |

**Table 8: Ablation experiment settings and FIDs. "Surrogate Model" refers to DM that employs accuracy-aware acceleration, but is not fine-tuned using feature alignment algorithm.**

| Ablation Settings | Setting Configurations | FID |
|---|---|---|
| Surrogate Model | No Training | 317.41 |
| SettingA | $\lambda_{\text{FA}}$=0.01 $\lambda_{\text{EWO}}$=1 $\lambda_{\text{VWO}}$=1 | **57.72** |
| SettingB | $\lambda_{\text{FA}}$=0.01 $\lambda_{\text{EWO}}$=1 $\lambda_{\text{VWO}}$=0 | 61.31 |
| SettingC | $\lambda_{\text{FA}}$=0.01 $\lambda_{\text{EWO}}$=0 $\lambda_{\text{VWO}}$=1 | 64.55 |
| SettingD | $\lambda_{\text{FA}}$=0 | 70.73 |

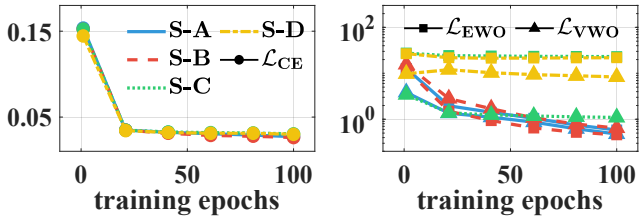



**Figure 6: Variation of losses during training. "S-X" denotes the ablation setting named "SettingX".**

## 5 Conclusion

In this paper, we present PrismDM, a novel MPC-based framework for accurate and efficient secure DM sampling. For the first time, we systematically identify severe image quality degradation in accelerated secure DM sampling and trace its root cause to the coupling between acceleration-induced bias and the iterative nature of DM sampling. To address this degradation, we propose an accuracy-aware acceleration scheme to mitigate acceleration-induced bias and a feature alignment algorithm to stabilize the sampling trajectory, thereby enabling secure, efficient, and accurate DM sampling. Experimental results demonstrate that PrismDM consistently and significantly improves both inference efficiency and sample quality across diverse and representative DMs and settings. Future work will aim to further enhance the accuracy and efficiency of MPC-based sampling, with an emphasis on both protocol-level optimization and system-level integration. As secure computation, acceleration strategies, and generative models continue to advance, secure DM sampling is expected to become increasingly practical across a wide range of real-world applications.

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